



A steel property optimization model based on the XGBoost algorithm and improved PSO

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ARTICLE INFO

Keywords:

Tensile strength

Plasticity

XGBoost

Particle swarm optimization

ABSTRACT

Exploring the relationships between the properties of steels and their compositions and manufacturing parameters is extremely crucial and indispensable to understanding the science of materials, and subsequently developing new materials. Tensile strength and plasticity, as two important properties of steels, are key to the improvement and optimization of the mechanical properties of steels. In the present paper, we propose an optimization model combining XGBoost algorithm with improved PSO to address the continuous multivariable optimization problem. The main goal is to determine the mapping functions between the tensile strength and plasticity and their influencing factors, based on a diversity of machine learning models such as Linear Regression, SVM, XGBoost, etc. After evaluating the performance these models, we then select the XGBoost model with highest accuracy as the mapping function, which has not been done in previous studies. Moreover, the determined mapping function serves as the fitness value of particle swarm optimization, after which the tensile strength and plasticity optimization with many variables is realized. Finally, the experimental results are analyzed theoretically, and proven to be effective and reliable.

1. Introduction

With the rapid development of industry, the demand for new materials is continuously increasing. However, while advancing material science and engineering, practitioners in this field have traditionally relied greatly on observations made from cleverly designed controlled experiments and sophisticated physics-based models to establish the desired relations [1]. As a result, new material development is very time-consuming, and the cost is equally great.

In order to accelerate the design and manufacture of advanced materials, machine learning is increasingly applied in the field of materials. Ceder et al. [2] proposed a structure prediction model based on machine learning, which mined the correlation of experimental data. They then used the model to effectively guide quantum mechanics technology to the stable crystal structure of materials. Also, Lu et al. [3] developed a target-driven method to predict undiscovered hybrid organic-inorganic perovskites (HOIPs) for photovoltaics. In addition, machine learning methods have also been applied to other fields of materials, such as bulk metallic glasses [4], thermoelectric materials [5], crystal structure [6], etc.

Steel, as an important material in modern industry, has also been studied thoroughly in recent years. However, it is quite difficult to improve the properties of steel, which are influenced by large number of factors, such as chemical compositions, process parameters, and microstructures [7]. It has been widely acknowledged that the main chemistry compositions of steel are highly varied, including iron (Fe), carbon (C), chromium (Cr), manganese (Mn), and other alloy elements [8]. In addition, the processing parameters of steel are also quite complex, including heating temperature, heating time [9], size of parts, and other parameters. The microstructure of steel, which is influenced by chemical compositions and process parameters, determines its properties.

In order to improve the properties of iron and steel, a growing number of scholars have developed machine learning models to study the relationship between material properties and their related factors. For example, Fang et al. [10] applied genetic algorithms (GA) and support vector regression (SVR) to predict the atmospheric corrosion of metallic materials such as zinc and steel. The results showed that the hybrid model is a promising and practical real-time corrosion tracking method for steel. Subsequently, Choi [11] established Artificial Neural

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<https://doi.org/10.1016/j.commsci.2019.109472>

Received 11 October 2019; Received in revised form 4 December 2019; Accepted 7 December 2019

Available online 18 December 2019

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Network (ANN) model to obtain the relationship of composition and processing parameters with ductility and impact toughness. Then, the multi-objective genetic algorithm was used to find the optimal solution of the various parameters. Additionally, Shun et al. proposed a machine-learning-based method using nonlinear programming for numerous properties of the materials, and solved the problem by using the Interior Point Algorithm [12].

However, these studies mainly developed a predictor for steel property, and the accuracy of these algorithms is low. And most methods have not solved the optimal combination of chemical compositions and process parameters to improve steel properties. Conversely, the exploration of optimal parameter combination is necessary for the production of steel materials with high mechanical properties in industry. Tensile strength [13] and plasticity, which respectively have the ability to resist fracture and deformation, are directly related to material life and mechanical reliability. Therefore, in this study we seek to establish machine learning model for predicting the tensile strength and plasticity of steel with 27 features, and use an improved Particle Swarm Optimization (PSO) [14] to optimize the values of these variables. In fact, the optimization analyses are very common in engineering application [15,16]. For instance, the AT algorithm [17] was applied to maximize the load attenuation and minimize the structural mass in paper [18]. In this study, we calculate the potential optimum chemical composition and process parameters by improved PSO because of its simplicity and excellent performance. The optimization results can provide appropriate guidance for the design of steel materials with high tensile strength and plasticity.

This paper is organized as follows. In Section 2, the key idea is to establish the mapping function between the tensile strength and plasticity and 27 features, including chemical element content and process parameters. We then compare 10 different machine learning models, and finally select the best XGBoost as our mapping function. In Section 3, the Improved PSO with sigmoid-like inertial weight is used to optimize the mapping function and obtain the optimal solution. In Section 4 and Section 5, we briefly discuss the results of this study and our future research. In Fig. 1, the workflow of creating machine learning model and optimizing these variables is described schematically.

2. Mapping function selection and analysis

In this section, we first analyze the relevance of 27 features and tensile strength and plasticity, then select XGBoost as our mapping function from ten different machine learning models. Finally, we rank the feature importance in order to solve the optimal solution more effectively, based on the improved PSO. More detail on these is elaborated in sequence in the following subsections.

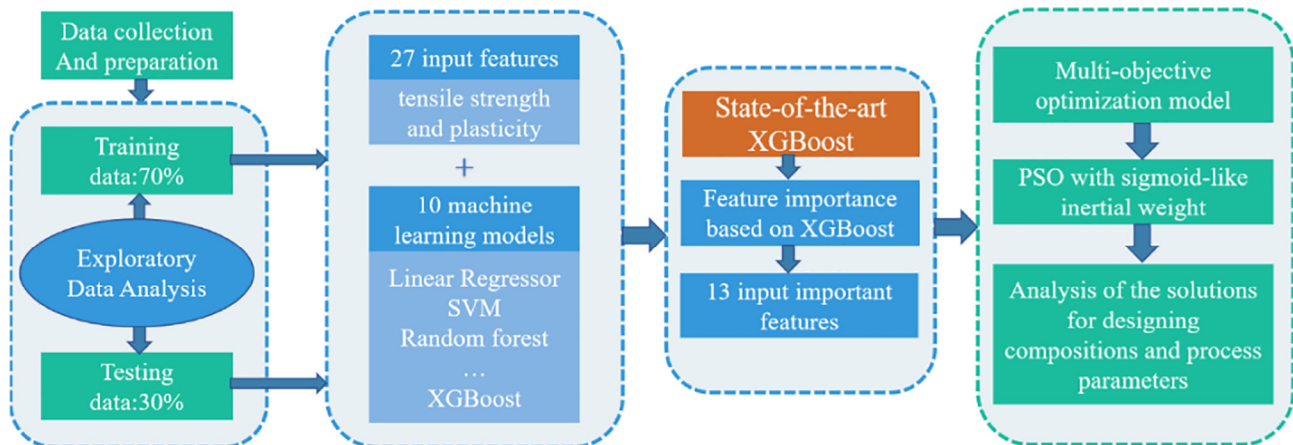


Fig. 1. Overall framework of the proposed approach to the tensile strength and plasticity optimization of steel.

Table 1
Data features.

Abbreviation	Details	Abbreviation	Details
Ft	Furnace temperature	Ac	Acid soluble Aluminum
Et	Exist temperature	Ti	Titanium
At	Annealing temperature	B	Boron
Th	Thickness	Sn	Tin
Wi	Width	As	Arsenic
S	Sulfur	Zr	Zirconium
Cu	Copper	Ca	Calcium
Ni	Nickel	Pb	Lead
Cr	Chromium	Ceq	Carbon equivalent #1
Mo	Molybdenum	PCM	Carbon equivalent #2
V	Vanadium	Sb	Antimony
Nb	Niobium	N	Nitrogen
Al	Aluminum	O	Oxygen
Wu	Tungsten		

2.1. Dataset

A data-driven model is the collection of data sets, and largely determines the quality of the model. In this study, our steel production dataset is taken from a previous paper [12]. There are 63,137 samples in the original data. As some samples lack necessary and important data, in the end 63,000 samples were chosen for our study. To effectively utilize these data, the dataset is divided into two parts. The first part is used for training the model, and the other for validating the accuracy of the model. The training set and test set respectively account for 70% and 30%. Each sample contains the following information: (1) the dataset has 27 features, as listed in Table 1, which include the chemical element content and process parameters; and (2) tensile strength and plasticity are the target properties for which we intend to build the mapping function and optimization model. The features of the data sets can be divided into the following two categories:

- Processing details: furnace temperature, existing temperature, annealing temperature, thickness, width.
- Chemical element content: %C, %S, %Mn, %Ni, %Cr, %Cu, %Mo, etc. (all in wt. %)

2.2. Data standardization

Standardization is the application of a mathematical transformation to each of the individual components of a multivariate representation, so that all of the variables make a comparable contribution to the result of the model [19]. Milligan et al. [20] summarized seven different (but often closely related) standardization methods. The most common and traditional standardization method is Z-Score, in which variables are

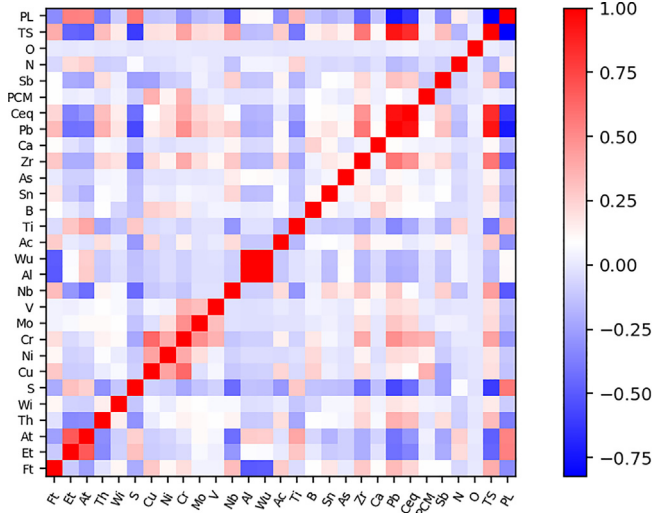


Fig. 2. Coefficient of correlation of 27 features and tensile strength (TS) and plasticity (PL).

converted to zero mean and unit variance. The method is defined as follows:

$$S = (X - \mu) / \sigma \quad (1)$$

where X represents the original value of the variables (features), and μ and σ respectively denote the mean and standard deviation of the observed value of the variables. Then, the values of these variables can be fed into the models as input parameters.

2.3. Xgboost algorithm

Before building the model, we perform a correlation analysis between variables. Fig. 2 shows the coefficient of correlation, which is an analysis of the relationship between independent variables (27 features) and target variables (tensile strength and plasticity). It can be seen intuitively that some features have a strong correlation with others. However, this is only a linear relationship analysis, which may not explain the interaction between features. Therefore, we require more complex prediction models, and many other machine learning models are discussed in the following sections.

XGBoost, a scalable tree boosting system, was proposed by the Chen et al. [21] and has been widely used in Kaggle's Higgs sub-signal recognition contest. More recently, it has attracted wide attention due to its outstanding efficiency and high prediction accuracy. In fact, XGBoost is an improved GBDT algorithm [22], which consists of many decision trees and is typically used in the field of classification and regression. However, XGBoost differs from GBDT in some respects. First, the GBDT algorithm only uses the first-order Taylor expansion, while XGBoost adds a second-order Taylor expansion to the loss function. Second, normalization [23] is utilized in the objective function to prevent over-fitting and reduce the complexity of the model. The structure of the XGBoost algorithm is shown in Fig. 3 and the details of the model are illustrated as follows.

Assume that a dataset is $D = \{(x_i, y_i) | i = 1, 2, \dots, n\}$, and the model with k trees is trained or learned. The result ($\hat{y}_i$) of the model is as follows:

$$\hat{y}_i = \phi(x_i) = \sum_{k=1}^K f_k(x_i), \quad f_k \in F \quad (2)$$

where F is the hypothesis space, and $f(x)$ is a regression tree:

$$F = \{f(x) = \omega_{q(x)}\} \quad (3)$$

In Formula (3), $q(x)$ is the leaf node of x -th sample and ω is the leaf

score. The following is the predicted result of the t -th iteration:

$$\hat{y}_i^t = \hat{y}_i^{t-1} + f_t(x_i) \quad (4)$$

Therefore, the objective function is:

$$J(f_t) = \sum_{i=1}^n L(y_i, \hat{y}_i^{t-1} + f_t(x_i)) + \Omega(f_t) \quad (5)$$

where L is the loss function and $\Omega(f_t)$ represents the complexity of the model. T and ω denote the number of leaf node and score, respectively.

$$\Omega(f_t) = \gamma \cdot T_t + \lambda \frac{1}{2} \sum_{j=1}^T \omega_j^2 \quad (6)$$

Next, Formula (6) is simplified by the second-order Taylor expansion:

$$J(f_t) = \sum_{i=1}^n [L(y_i, \hat{y}_i^{t-1}) + g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (7)$$

$$g_i = \frac{\partial L(y_i, \hat{y}_i^{t-1})}{\partial \hat{y}_i^{t-1}} \quad (8)$$

$$h_i = \frac{\partial^2 L(y_i, \hat{y}_i^{t-1})}{\partial \hat{y}_i^{t-1}^2} \quad (9)$$

According to the analysis above, the final objective function is as follows:

$$J(f_t) = \sum_{i=1}^n \left[g_i \omega_{q(x_i)} + \frac{1}{2} h_i \omega_{q(x_i)}^2 \right] + \gamma T + \lambda \frac{1}{2} \sum_{j=1}^T \omega_j^2 \quad (10)$$

Finally, the objective function is optimized, and the optimal solution is:

$$\omega_j^* = - \frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \quad (11)$$

$$J(f_t) = - \frac{1}{2} \sum_{j=1}^T \frac{\left(\sum_{i \in I_j} g_i \right)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T \quad (12)$$

2.4. Comparison of different models

Without loss of generality, we establish other machine learning models, such as Linear Regressor [24], K Nearest Neighbor [25], Support Vector Machine [26], Decision Tree [27] and Random Forest [28]. Table 2 provides brief descriptions of the models.

To evaluate the performance of these models, a multi-criteria assessment with various goodness-of-fit statistics is performed using all of the data vectors to test the accuracy of the trained models [29]. The three metrics are used for the model evaluation [30]. Supposing that $y_1, y_2, \dots, y_n$ are actual values, $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$ are predicted values, and $\bar{y}$ is the mean of y_i , the formulas are defined as:

$$R^2 = \frac{\sum_{i=1}^n (y^{(i)} - \bar{y})^2}{\sum_{i=1}^n (y^{(i)} - \bar{y})^2} \quad (13)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y^{(i)} - \bar{y}| \quad (14)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y^{(i)} - \bar{y})^2} \quad (15)$$

where R^2 is used to measure the approximation degree of data to fitting values, MAE represents the absolute error values between the predicted and actual values, and RMSE is the mean squared error.

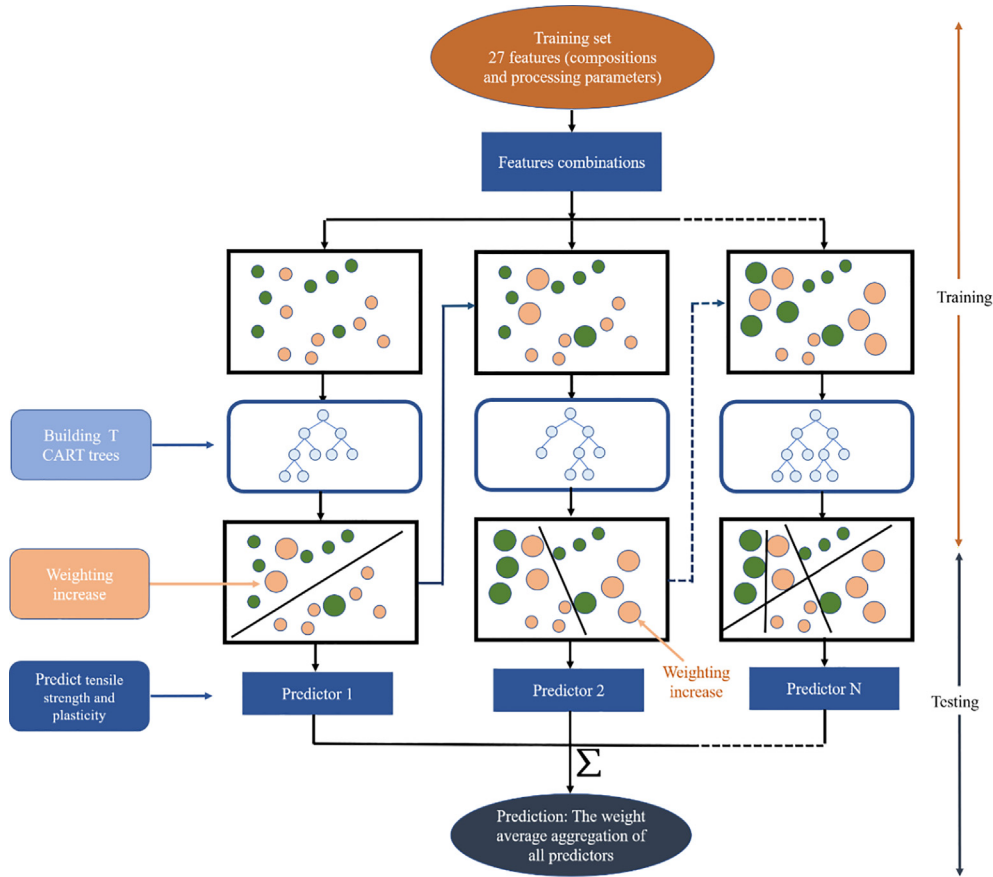


Fig. 3. The structure of the XGBoost algorithm.

Table 2
Ten different models to build the mapping function.

Models	Note
LR	A model of Linear Regressor
KNN	A model of K Nearest Neighbor
Linear-SVM	Linear Support Vector Machine
Poly-SVM	Polynomial Support Vector Machine
RBF-SVM	RBF Support Vector Machine
DT	Decision Tree Regressor
GB	Gradient Boosting Regressor
ET	Extra Trees Regressor
RF	Random Forest Regressor
XGBoost	eXtreme Gradient Boosting

Table 3
Comparison of different TS-predicted models.

Algorithm	R^2	MAE	RMSE	Train Time (s)	Test Time (s)
Linear-SVM	0.9348	21.3883	31.2845	1458.1808	11.2909
LR	0.9354	21.7026	31.1468	0.0690	0.0020
DT	0.9382	19.7672	30.4565	1.2635	0.0200
GB	0.9534	17.6755	26.4473	10.5951	0.1235
Poly-SVM	0.9557	16.7902	25.7890	559.9789	35.4503
KNN	0.9573	16.3187	25.3125	7.9982	40.5517
ETR	0.9607	15.4664	24.2955	3.9547	0.0800
RBF-SVM	0.9617	15.3627	23.9831	146.6859	14.5473
RF	0.9619	15.4076	23.9262	7.5742	0.0700
XGBoost	0.9628	15.1607	23.6432	7.3238	0.0460

In our experiments, RF, SVM, GD, etc. are implemented using the scikit-learn library the xgboost library in Python. The test platform includes the laptop equipped with Core i5- 4210H CPU and 8G RAM.

Table 3 presents the prediction accuracy, train time and test time of the 10 mapping functions used in the work in terms of the tensile strength, and Fig. 4 shows the distribution of R^2 , MAE RMSE for these models. It is clearly observed that seven models can achieve a high predictive accuracy, with R^2 values approximately over 0.95. From the results, we can also observe that three of the models (RF, RBF-SVM and XGBoost) perform better than the others, with R^2 values exceeding 0.96, in particular XGBoost ($R^2 = 0.962$, Train time = 7.3238 s, test time = 0.0460 s). However, the accuracy of the three linear models (LR, poly-SVM and linear-SVM) is poor, their values all being below 0.94. This is also in line with the actual situation of the project. Due to the fact that the relationship between the tensile strength and 27 features is obviously nonlinear, the accuracy of the other three models (KNN, GB and DT) lie in the middle of the models.

Similarly, the selection of plasticity model is the same, as it is vividly depicted from Table 4 and Fig. 5 that the accuracy of RF and XGBoost is higher than the others. Also, the XGBoost model has the lowest MAE and RMSE among all models, at 2.79 and 3.83, respectively. Meanwhile, the train time and test time of XGBoost are both very short, with 7.0596 s and 0.0479 s respectively. In summary, the XGBoost model has the greatest capability in predicting the objective values and thus is determined as the mapping function between the features and tensile strength and plasticity.

In order to illustrate the fitting effect of XGBoost, Fig. 6 presents the distribution of the actual and predicting tensile strength and plasticity in the test data. The closer the scatter points are to a straight line, the stronger the prediction ability will be. It is directly perceived that the fitting effect of the model is desirable and acceptable.

According to the analysis mentioned above, we can clearly observe that the XGBoost model has the highest value of R^2 , and relatively low train time, test time, MAE and RMSE values. Overall, the XGBoost

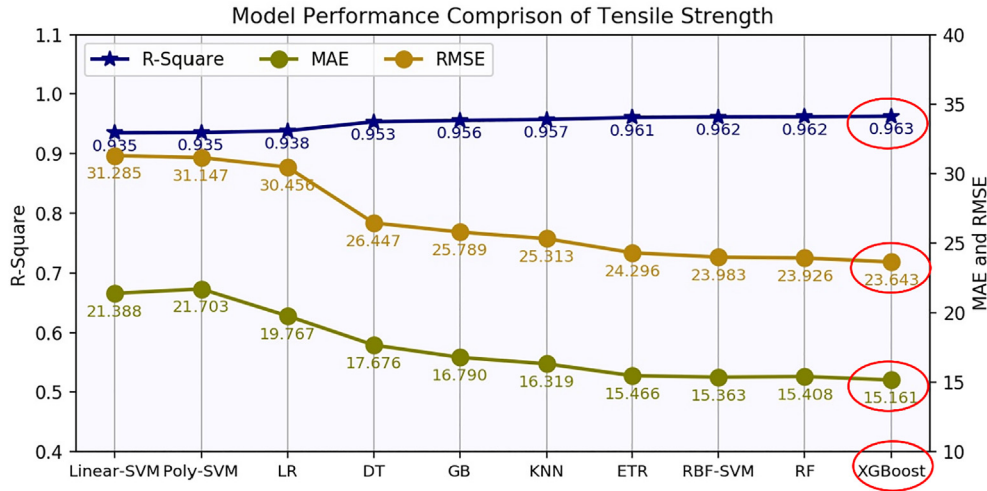


Fig. 4. Results of TS-predicted models: R^2 , MAE and RMSE for ten models. The model with higher R^2 values and lower MAE and RMSE values represents higher fitting accuracy.

Table 4
Comparison of different PL-predicted models.

Algorithm	R^2	MAE	RMSE	Train Time (s)	Test Time (s)
DT	0.6517	3.8247	5.3535	1.1133	0.0090
Linear-SVM	0.7207	3.5633	4.7939	835.6601	16.6842
LR	0.7224	3.5803	4.7794	0.0719	0.0020
GB	0.7820	3.1226	4.2352	10.5700	0.0260
Poly-SVM	0.7849	3.0501	4.2072	500.8096	19.4753
KNN	0.7886	3.0158	4.1707	8.3701	41.1148
RBF-SVM	0.7926	2.9895	4.1313	174.1398	26.3103
ETR	0.7945	2.9748	4.1120	3.6061	0.0730
RF	0.8038	2.9155	4.0178	7.1583	0.0700
XGBoost	0.8219	2.7938	3.8282	7.0596	0.0479

model is more efficient than the other machine learning models. Accordingly, it is selected as the final mapping function and optimization objective of the PSO.

2.5. Feature importance ranking

To further study the influence of a single feature on the tensile strength and plasticity, we calculate the relative importance [31] of the 27 features based on the XGBoost model that we selected. The idea

behind this method is to use the machine learning algorithm we built to build a prediction model for each individual feature. Generally speaking, a single feature is run on the XGBoost model of the data samples obtained using the bootstrapping method [32]. Then, the importance of every feature is ranked on the basis of the scores.

Fig. 7 shows the importance of the 27 attributes to tensile strength and plasticity. Clearly, these features (Ceq, Ti and Pcm) make a great difference in the tensile strength, while the others (Sn, Pb and W) are not as important. The results of our model also agree well with industrial reality. For instance, two types of carbon equivalence (Ceq and Pcm) [33], which consist of with large numbers of alloying elements, have a great impact on the tensile strength of steel. Additionally, Niobium (Nb), a very critical chemical element, as it contributes to the strength of steel through refined grain size strengthening and solid solution strengthening [12]. In contrast, some chemical element such as Arsenic (As) and Lead (Pb) are relatively trivial, and can generally be neglected. Therefore, in order to reduce the complexity of the optimization model, we choose the features of the top 10 as the variables for tensile strength. For plasticity, the top four (Ceq, Pcm, Ti and Nb) are consistent with tensile strength. However, the existing temperature (Et) and thickness (Th) of part are also important for the plasticity. Furthermore, it is noted that the moderate Nitrogen (N) content can improve the plasticity of steel. Finally, taking the top 10 variables of the

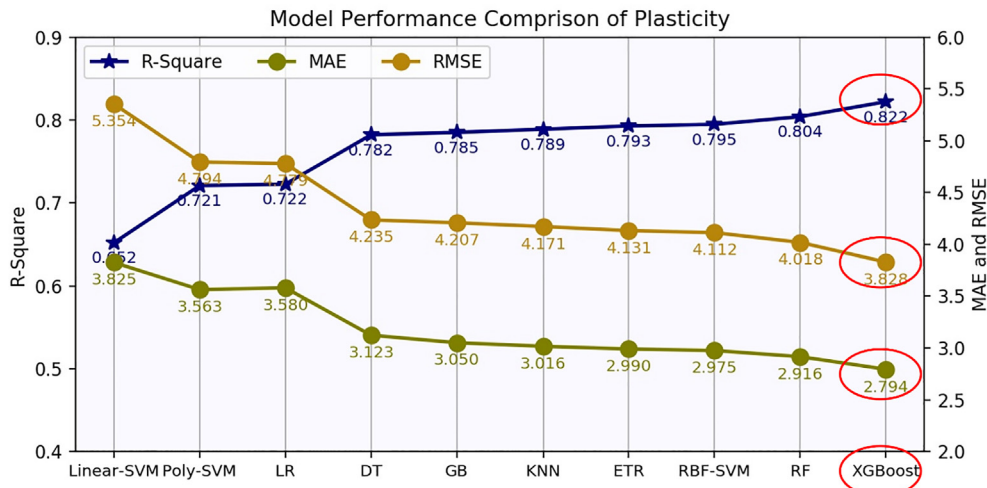


Fig. 5. Results of PL-predicted models: R^2 , MAE and RMSE for ten models. The model with higher R^2 -values and lower MAE and RMSE values represents higher fitting accuracy.

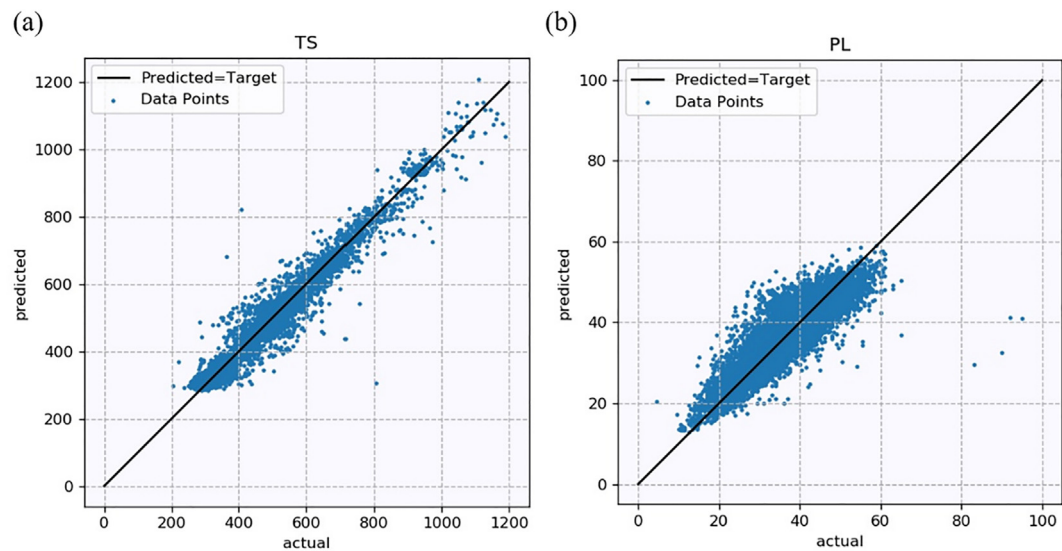


Fig. 6. Predicted versus actual tensile strength (TS) and plasticity (PL) (in MP) of the XGBoost model. The X-axis and Y-axis respectively represent the actual and predicted values.

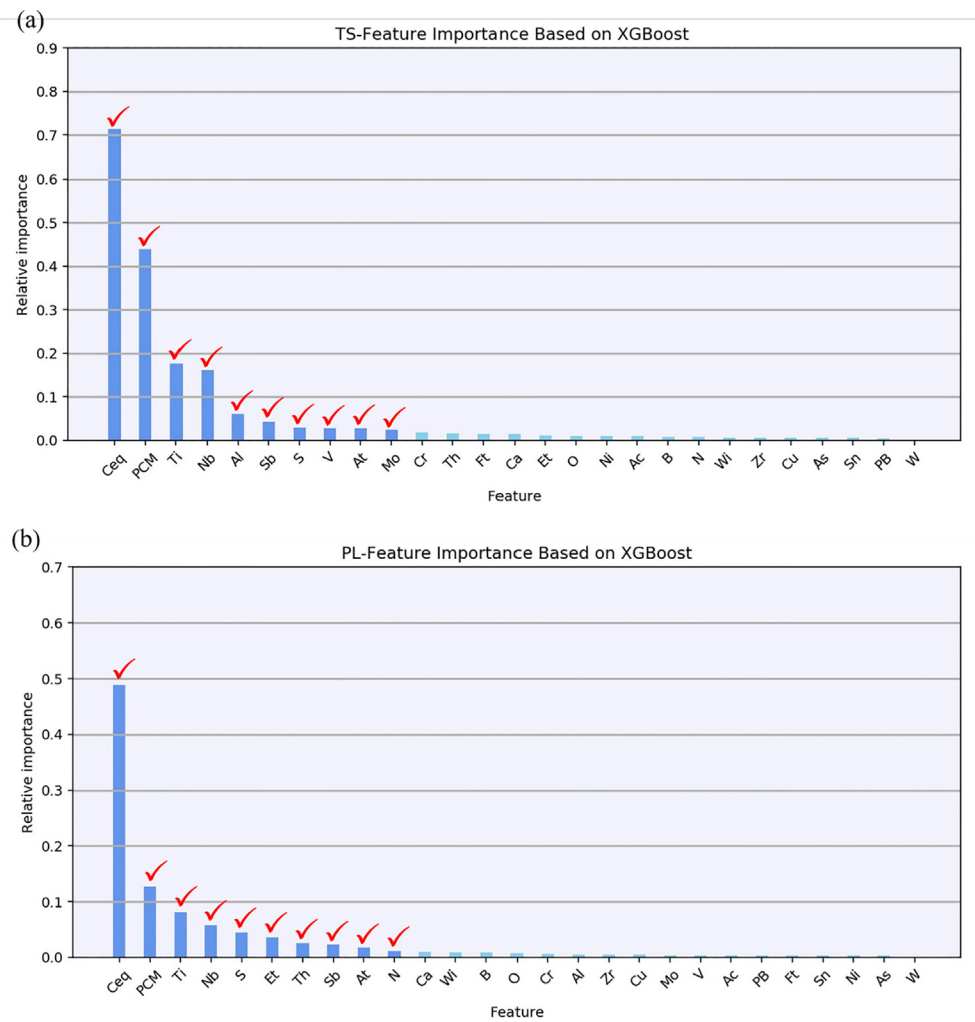


Fig. 7. The importance ranking of 27 input features. The X-axis and Y-axis respectively denote the input features and importance of each feature to tensile strength (TS) and plasticity (PL).

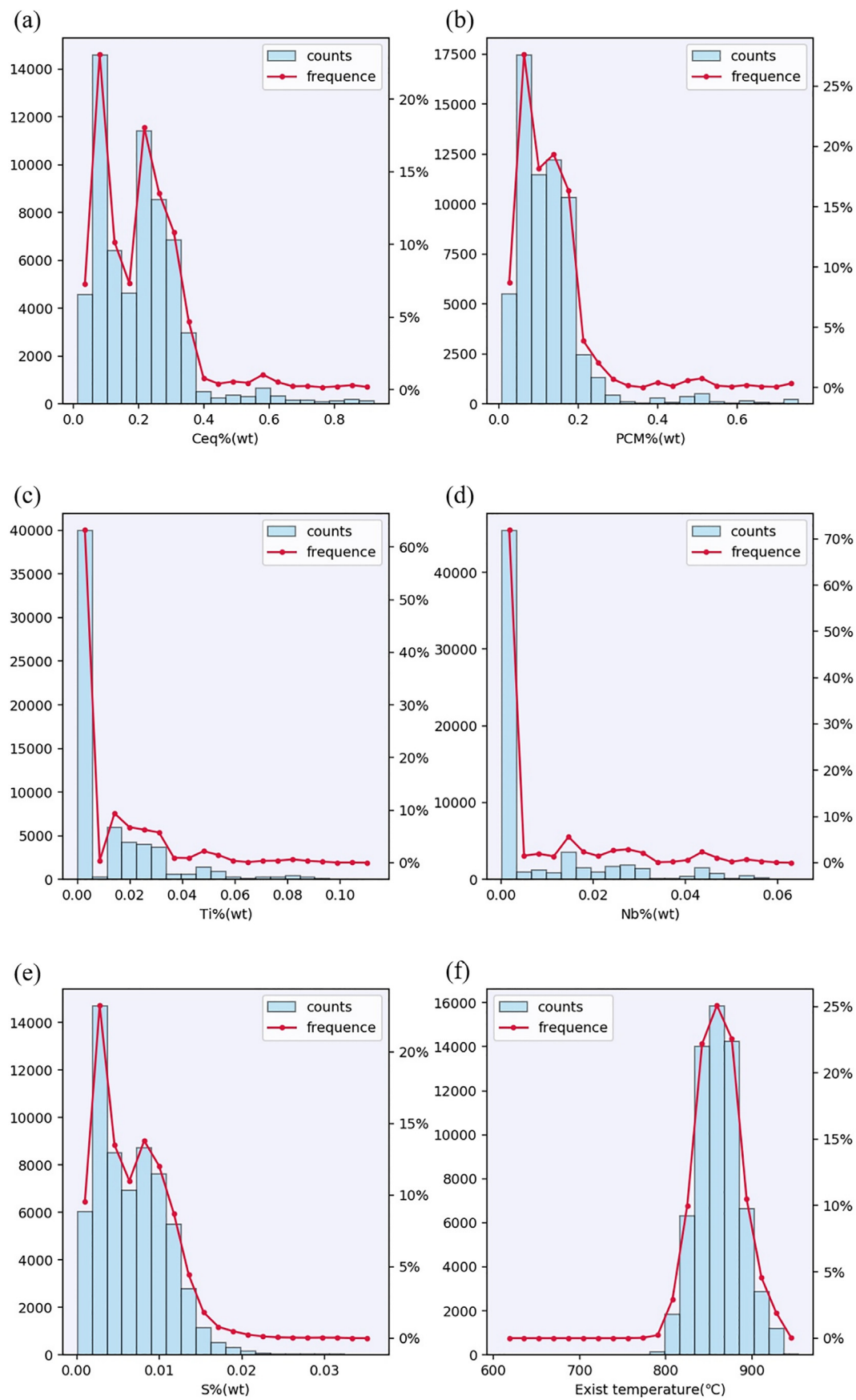


Fig. 8. Histograms of 13 important chemical compositions and process parameters in the data set.

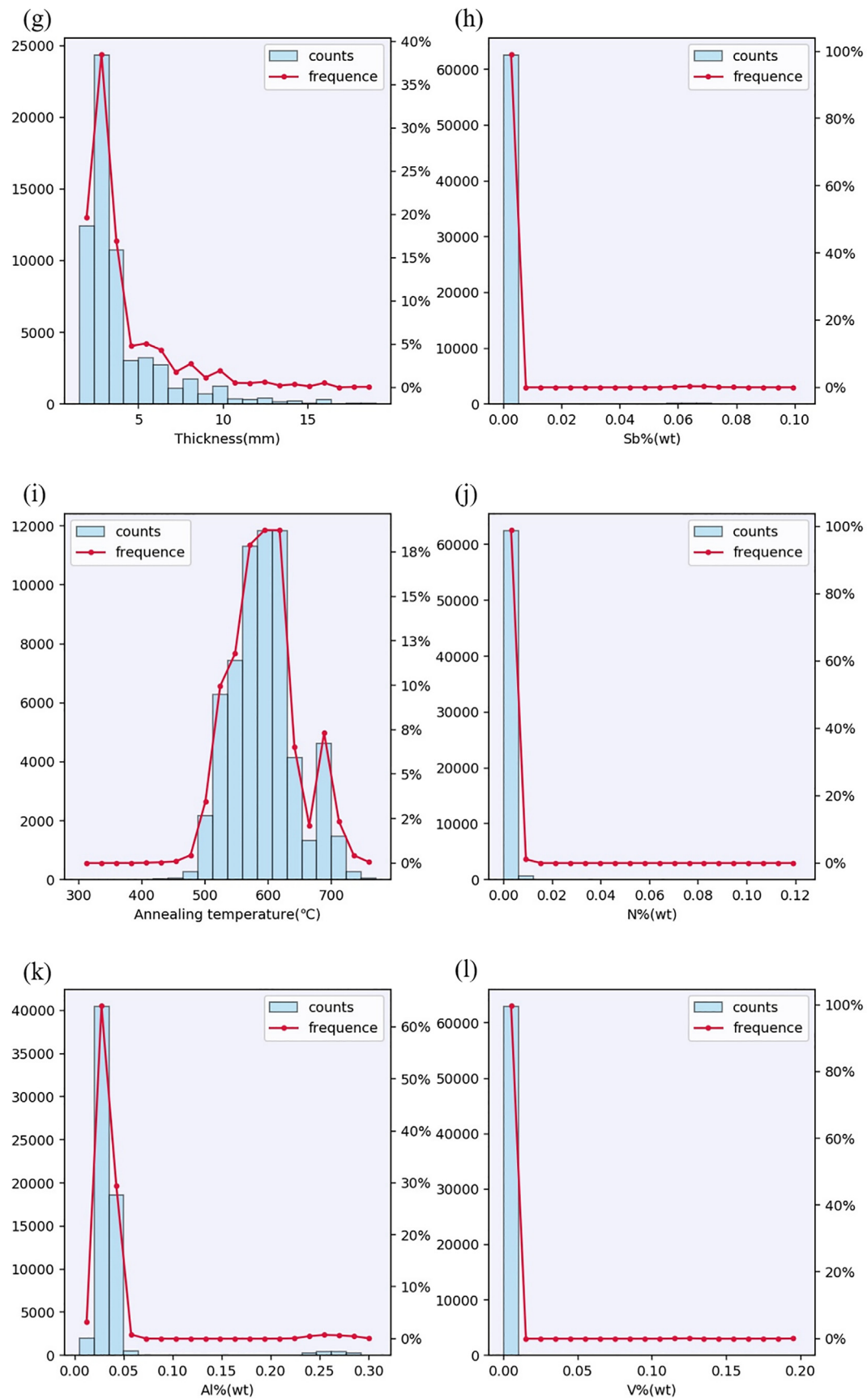


Fig. 8. (continued)

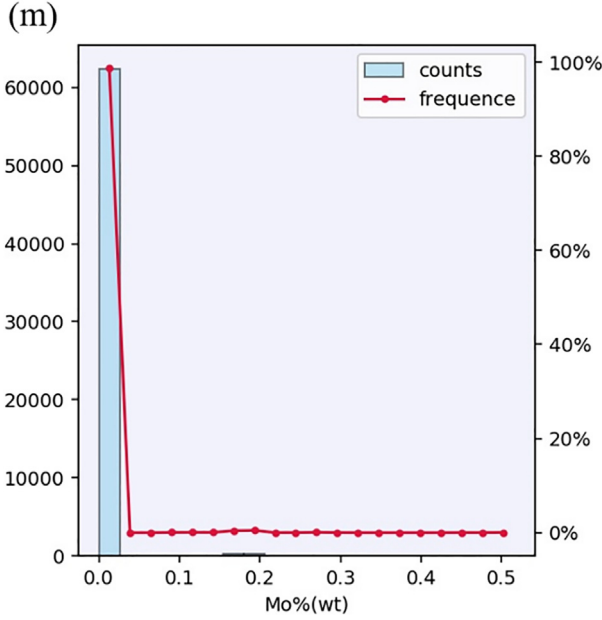


Fig. 8. (continued)

tensile strength and plasticity into consideration, 13 variables then become the later optimization objective.

Fig. 8 depicts the distributions of 13 important chemistry elements and manufacture parameters in the data set, which can lay a good foundation for our later optimization.

3. Solution of the model based on improved PSO

In this section, the two mapping functions about tensile strength and plasticity are used as objective functions and a multi-objective model is established. In order to solve the problem conveniently, multi-objective optimization is transformed into single-objective optimization by weighting. Then, the final objective function is acted as the fitness value of PSO with sigmoid-like inertial weight. More details are as follows.

3.1. Establishment of model

According to the work mentioned above, we can select the XGBoost model as our mapping function. A total of 13 important features have been also selected as variables that need to be optimized. $X = [X_1, X_2, X_3, \dots, X_n]$ ($n = 13$) denotes a series of values of the chemical element content and process parameters of steel, Y represents the tensile strength or plasticity, and $f(X)$ is the mapping function from X to Y .

$$f(X) = \sum_{i=1}^p \omega_i f_i(X) \quad (p = 2) \quad (16)$$

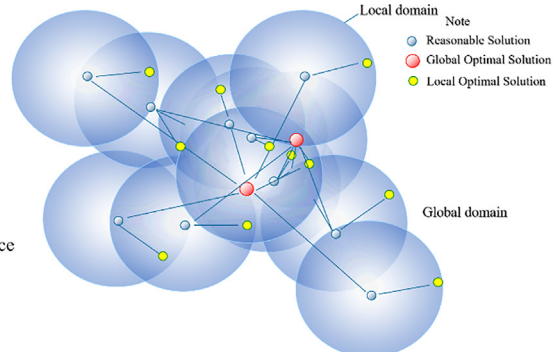
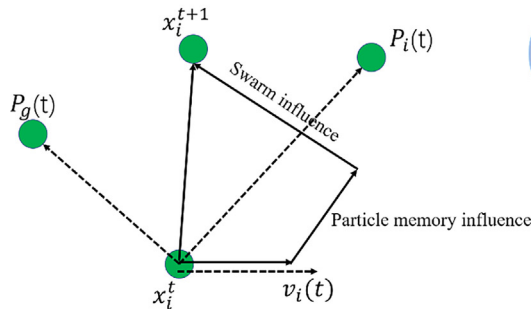


Fig. 9. Graphical representation of particle evolution and search for optimal solutions.

where $f_1(X)$ and $f_2(X)$ are the mapping function of the tensile strength and plasticity based on the XGBoost model, and ω_i is the weight coefficient ($\omega_i = 1$).

Therefore, the optimization model can be defined as follows:

$$\begin{cases} \max(f(X)) \\ X_{1min} \leq X_1 \leq X_{1max}, \\ X_{2min} \leq X_2 \leq X_{2max}, \\ \vdots \\ X_{nmin} \leq X_n \leq X_{nmax} \quad (n = 13). \end{cases} \quad (17)$$

where the X_{nmax} and X_{nmin} are the upper and lower boundaries according to the original dataset. The content of each element is limited by its actual maximum and minimum content in iron. Therefore, each X_n is between the maximum (X_{nmin}) and minimum (X_{nmax}). Meanwhile, the process parameters possess boundaries due to the influence of actual technology.

3.2. Solution of the model

The determination of optimum chemical compositions and process parameters is a combinatorial optimization problem, and is typically realized by applying optimization algorithms. At present, the evolutionary optimization algorithms are widely used in engineering issues, due to their excellent performance, such as in genetic algorithms (GA) [11], ant colony optimization (ACO) [34], simulated annealing (SA) [35], neural networks [36], firefly algorithm (FA) [37,38], cuckoo search (CS) [39,40], etc. However, unlike these evolutionary algorithms (EAs) [41], the PSO algorithm is motivated by the simulation of social behavior instead of survival of the fittest. Compared with other EAs, it is clear that PSO has the advantages of fewer control parameters, better convergence, and easy implementation [42]. Therefore, in the present work, PSO is used to optimize the chemical compositions and process parameters.

3.2.1. Introduction to the standard PSO

Particle swarm optimization (PSO) was first proposed by Kennedy and Eberhart [43] in 1995, and its concept originated from the study of bird prey behavior. These particle swarms consist of volume-less particles with random velocities, each of which represents a feasible solution. The optimal solution is determined by moving the particles in the solution space. Fig. 9 describes the process of determining the optimal solution for particles.

Given a swarm of m particles, the position and velocity of one particle is represented as $X_i = (x_{i1}, x_{i2}, \dots, x_{iD})$ and $V_i = (v_{i1}, v_{i2}, \dots, v_{iD})$, respectively, and D is the dimensions of the solution space. In the process of particle search, the global best position ($P_g = (p_{g1}, p_{g2}, \dots, p_{gD})$) in the group and local best position ($P_i = (p_{i1}, p_{i2}, \dots, p_{iD})$) of each particle are both recorded. The velocity and position of the particle are updated according to the following equation:

$$v_i^{t+1} = \omega v_i^t + c_1 r_1 (p_i^t - x_i^t) + c_2 r_2 (p_g^t - x_i^t) \quad (18)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \quad (19)$$

where ω is the inertia weight, c_1 and c_2 are acceleration coefficients, usually $c_1 = c_2 = 1$, and r_1 and r_2 are the two random numbers uniformly distributed in the range (0,1).

The particle velocity update formula consists of three parts: The first part is the particle's previous velocity, and the second is the "cognitive" part, which expresses the thinking of the particle itself. This can be interpreted as the distance between the current position of particle and its best position. The third part is the "society" part, which represents the information sharing and cooperation among particles. It is the distance between the current position of particle and the best position of the group.

3.2.2. Improved PSO with sigmoid-like inertial weight

It is known that the overall performance of PSO is strongly affected by the inertial weight [44]. The inertial weight (ω) determines the ability of global search and local search, which is very crucial for particles to find the optimal solution. When the inertial weight is set as a constant (such as 0.6) the first time, then the results are not satisfactory due to the failure to balance the exploration and exploitation. As a result, a large number of inertial weight strategies have been proposed to solve this problem. First, in Ref. [45], a linear decreasing weight (Eq. 20) was developed. Then, we also observed that a relatively large inertial weight value can promote the global search, while a small value can increase the ability of the global search. Therefore, in order to solve this problem, Cui et al. [46] proposed an inertial weight strategy (Eq. 21) that first increased, then decreased. The value of the inertial weight fluctuates between 0.4 and 0.9.

Subsequently, Chen et al. [47] devised a nonlinear decreasing strategy (Eq. 22) to update the inertial weight. They observed that the performance of the concave function based on decreasing inertial weight was better than the linear function in many continuous optimization problems.

Eventually, in order to combine the linear inertial weight and nonlinear inertial weight, a novel sigmoid-like inertial weight [42] was constructed to balance the ability of exploration (global search) and exploitation (local search). The widely used activation function is the sigmoid ($y = 1/(1 + \exp(-x))$) in deep learning, which can accurately achieve the tradeoff between linear and nonlinear behavior, as shown in Fig. 9. On the basis of the sigmoid, a novel and effective sigmoid-like inertia weight is proposed in Eq. 23, as shown in Table 5.

Table 5 summarizes the formulas of the four different inertial weights. In these methods, t and t_{max} respectively denote the current iteration and maximum number of iterations. Meanwhile, ω_{max} and ω_{min} represent the initial and final value of the inertial weight. In order to display the trend of change of the inertial weight, their iterative curves are presented in Fig. 10.

Table 5

Summary of four linear and nonlinear inertial weights.

Equation	Different inertial weights	Related description
Eq. 20	$\omega_0(t) = \frac{t_{max}-t}{t_{max}}(\omega_{max} - \omega_{min}) + \omega_{min}$	Linear decrease
Eq. 21	$\omega_1(t) = \begin{cases} 1 \times \frac{t}{t_{max}} + 0.4, & (0 \leq \frac{t}{t_{max}} \leq 0.5) \\ -1 \times \frac{t}{t_{max}} + 1.4, & (0.5 \leq \frac{t}{t_{max}} \leq 1) \end{cases}$	First increase, then decrease
Eq. 22	$\omega_2(t) = -(\omega_{start} - \omega_{end})\left(\frac{t}{t_{max}}\right)^2 + \omega_{start}$	Convex function
Eq. 23	$\omega_3(t) = \begin{cases} 0.9, & (t \leq \alpha t_{max}, \alpha = 0.2) \\ \frac{1}{1 + e^{(10t-2t_{max})/t_{max}}} + 0.4, & (otherwise) \end{cases}$	Sigmoid-like function

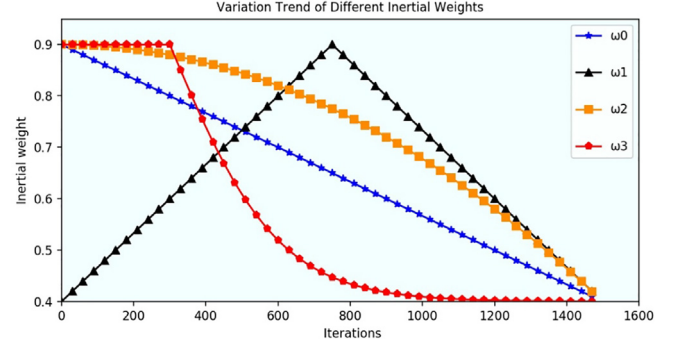


Fig. 10. Curves of different inertial weights.

3.3. Experiments of tensile strength and plasticity optimization with the improved PSO

Based on these different inertial weight strategies, the chemical compositions and process parameters (13 important features) are optimized using PSO algorithms. First, some hyperparameters of PSO should be set properly, such as two positive constants (c_1 and c_2), population size and iteration times, etc. Second, the initial position of particles is selected from the data set, so that the optimal solution can be searched quickly in the search space. Aside from this, it is noted that the chemical element and process parameters are limited in the bounds due to the actual industrial situation. Therefore, the constraints must be taken into account in order to obtain more precise solutions. Ultimately, the objective function, the mapping function or XGBoost model, is used to calculate the fitness value of the PSO. In order to better understand the improved PSO algorithm described above, we provide a concise yet complete pseudocode to illustrate the optimization process. The pseudocode algorithm is elaborated as follows:

Algorithm: pseudocode of 13 important features optimization with PSO

1. Begin
2. Initialize the particle by randomly choosing the variable values of the datasets
3. While (the number of iterations is not reached)
 - Calculate the fitness (trained XGBoost function) of particle
 - For $m=1$ to the sum of particles
 - Find the best position of each particle
 - Find the best position of all particles
 - For $D=1$ to the dimensions of particle
 - Update the velocity of particles according to Equation (6)
 - Update the position of particles according to Equation (7)
 - Next D
 - Next m
 - Update the inertial weight using the linear or nonlinear function
 - Next iteration until the maximum iteration
4. End

4. Results and discussion

We conduct the optimization experiments based on the ω_0 - PSO, ω_1 - PSO, ω_2 - PSO and ω_3 - PSO. It is noted that the four methods denote linear decreasing inertia weight, first increasing then decreasing inertia weight, concave function decreasing based inertia weight, and the sigmoid-like inertia weight. To reveal the effectiveness of these methods, the curve of fitness value (tensile strength and plasticity) varying with iterations can be obtained. From Fig. 11, it is easy to distinguish that the ω_3 - PSO evolves faster, and quickly converges to the best solution compared with its counterparts. We can also clearly observe that the ω_0 - PSO and ω_2 - PSO evolve extremely slowly for the first 400 iterations, and tend to reach the local optimum with great speed. Therefore, the results of ω_1 - PSO and ω_2 - PSO are not satisfactory in terms of its speed and local convergence. In addition, it is

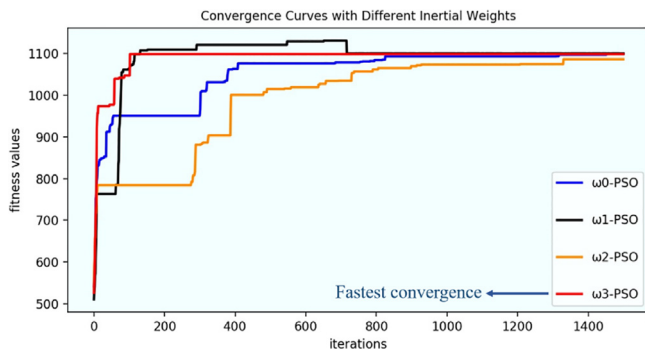


Fig. 11. Iterative curve diagrams of four different inertial weights.

Table 6

Statistical results of PSO with different inertial weights.

PSO algorithm	Convergence rate	Best solution	Average solution	Standard deviation
ω_0 - PSO	25/30	1129.284	1045.319	96.88784
ω_1 - PSO	26/30	1110.748	1039.144	85.31715
ω_2 - PSO	24/30	1123.093	1044.959	92.81041
ω_3 - PSO	28/30	1122.394	1040.373	81.03749

noted that the result of ω_1 - PSO is more effective with fast convergence than those of ω_0 - PSO and ω_2 - PSO. Nevertheless, the PSO is not so stable. Therefore, we conduct 30 independent runs, with each run being about 1,500 iterations. As shown in Table 6, the sigmoid-like inertial weight has the highest success rate, and its standard deviation is also relatively small, though its best solution and average solution are not very ideal. Fig. 12 shows the distribution of results of PSO with different inertial weights. It is clear that the optimization performance and stability of PSO with sigmoid-like inertial weight (ω_3 - PSO) are superior to those of the others.

Table 7 shows one of the optimal combinations of 13 chemical compositions and process parameters based on PSO with Sigmoid-like function. We can explicitly observe that carbon equivalent (Ceq) accounts for the largest proportion with about 0.878%, which indicates that carbon equivalent makes a great difference in the properties of steels, especially the tensile strength and plasticity. Carbon equivalent consists of a wide array of alloying elements, such as molybdenum content, chromium content, nickel content, copper content, etc. These chemical elements are essential and imperative for the properties of steels. Another carbon equivalent (PCM) that is the combination of silicon, molybdenum, vanadium, copper, chromium and boron represents approximately 0.740%. The two types of carbon equivalent are very important for the properties of steel. In addition, the titanium content (Ti) is commonly regarded as one of the essential chemical compositions that can significantly improve strength, hardness and wear resistance, while reducing plasticity and toughness. The experiment results validate this fact. Therefore, increasing the content of Titanium content is of great importance to improving the tensile strength of steel in the actual industrial production process. Besides, the heat treatment processes are equally important to the properties of steel such as annealing temperature (At) and exist temperature (Et). In view of this, we should attach importance to the adjustment of these two different temperatures, so as to enhance the properties of the materials. Eventually, the size of the parts such as thickness (Th) also affects the properties of the steel. In essence, only when all connected factors are fully considered can we obtain satisfactory tensile strength and plasticity.

5. Conclusions

Machine learning, based on statistics and big data, is quickly

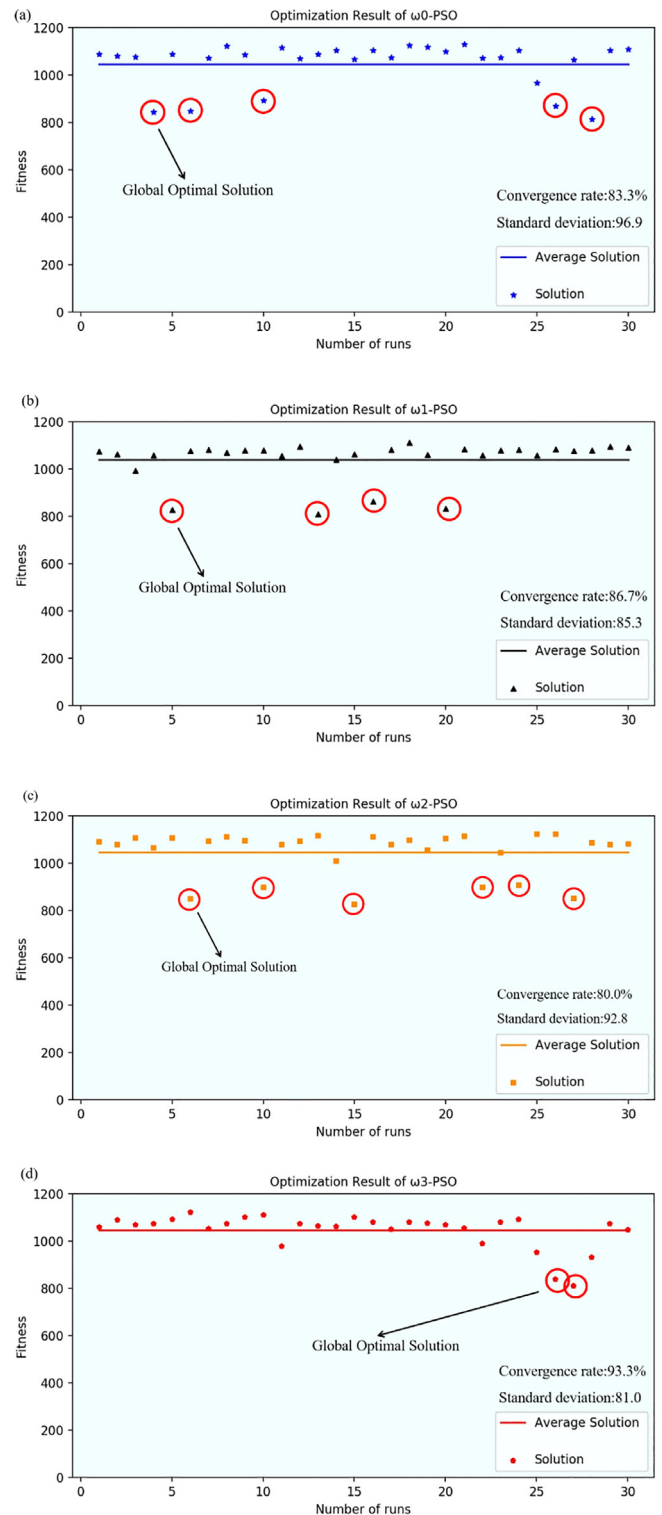


Fig. 12. Optimization results comparison of PSO with different inertial weights.

emerging to reduce the development time of new and improved materials, and thereby promote the progress of big data in the material industry. Most previous research on steel material has tended to focus on performing large numbers of experiments to improve the properties of steels, which wastes time and funds. However, in the present study, we utilize the machine learning and intelligent optimization algorithm to improve the tensile strength and plasticity of steel via industrial big data. First, we build 10 machine learning models to fit the connection between chemical compositions and process parameters and tensile

Table 7

One of optimization results of PSO based on sigmoid-like inertia weight.

Ceq	PCM	Ti	Nb	S	Et (°C)	Th (mm)	Sb
0.878188	0.739867	0.000734	0.045328	0.032806	857	8.749383	0.011243
At(°C)	N	Al	V	Mo	TS + PL(MP)	TS(MP)	PL(MP)
578	0.013699	0.030869	0.020032	0.034299	1099.443	1035.065	26.30703

strength and plasticity of steel. Second, through comparing the precision of these models, the XGBoost model is selected to be the mapping function. Subsequently, the top 13 chemical compositions and process parameters are identified as the optimization variables in the light of the importance to the model. Finally, an optimal combination of 13 chemical compositions and process parameters is obtained using the improved particle swarm optimization. Therefore, the model we established can provide guidance for steel production.

The data-driven and optimization models involve several application fields. First and foremost, the predictive model for tensile strength and plasticity of steels can also be applied to other properties of materials through machine learning. The entire process depends on large amounts of industrial data, without doing experiments trial and error. Furthermore, we can acquire a set of optimization chemical compositions and process parameters to improve the mechanical properties of steels. In conclusion, the model described in this study can provide some useful information for industrial iron and steel production and other materials.

At the same time, there are still some issues that need to be resolved. For example, we only studied the tensile strength and plasticity of steels, and ignored the other properties. These flaws will be the focus of our research in the future.

Author contribution

Kai Song: Manuscript preparation, Manuscript editing, Manuscript revision/review. Feng Yan: Guarantor of integrity of entire study, Study concepts, Study design. Ting Ding: Literature research, experimental studies. Liang Gao: Data acquisition, Data analysis/interpretation. Songbao Lu: Statistical analysis.

CRediT authorship contribution statement

Kai Song: : Conceptualization, Methodology, Software. **Feng Yan:** Data curation, Writing - original draft, Formal analysis. **Ting Ding:** Visualization, Investigation. **Liang Gao:** Writing - review & editing, Supervision, Validation. **Songbao Lu:** Resources, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are very grateful for the raw data used in a previous paper [12] by Shun Guo et al. The study is supported by the project of the National Key R&D Program "Joint Research on Advanced Technology and Application of Electric Vehicles based on China-US Cooperation" (2016YFE0102200). It is also supported by the National Natural Science Foundation of China (grant number U1864207) and National Youth Foundation of China (grant number 51605155).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.commatsci.2019.109472>.

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